



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.NOS.1

Divide Mixed Numbers

Lesson 6-10 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can divide mixed numbers by first changing them to improper fractions.

Language Objective: I can explain my steps using the words mixed number, improper fraction, convert, reciprocal, and simplify.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Divide Mixed Numbers

Mixed Number

Improper Fraction

Convert

Simplify

Reciprocal



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— Change mixed numbers to improper fractions, then multiply by the reciprocal.

2

A number with a whole number and a fraction, like $2\frac{1}{3}$, is a

.

3

A fraction whose numerator is greater than or equal to its denominator, like $\frac{7}{3}$, is an .

4

Changing a mixed number into an improper fraction is to

it.

5 Writing a fraction in its smallest equal form is to it.

6 To divide by a fraction, I multiply by its .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How does dividing mixed numbers help the detective?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Divide Mixed Numbers** — Change mixed numbers to improper fractions, then multiply by the reciprocal.
Dividir números mixtos — Cambiar los números mixtos a fracciones impropias y luego multiplicar por el recíproco.

☐ **Mixed Number** — A whole number plus a fraction, like $2 \frac{1}{3}$.
Número mixto — Un número entero más una fracción, como $2 \frac{1}{3}$.

☐ **Improper Fraction** — A fraction where the top is bigger than or equal to the bottom, like $\frac{7}{4}$.
Fracción impropia — Una fracción donde el número de arriba es mayor o igual al de abajo, como $\frac{7}{4}$.

☐ **Convert** — To change a number to a new form but keep the same value.
Convertir — Cambiar un número a otra forma sin cambiar su valor.

☐ **Simplify** — To make a fraction smaller using the same parts, like $\frac{2}{4} = \frac{1}{2}$.
Simplificar — Hacer una fracción más pequeña con las mismas partes, como $\frac{2}{4} = \frac{1}{2}$.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

**The detective can cut ____ pieces because $5 \frac{1}{4} \div \frac{3}{4} = 21/4 \times 4/3 = 84/12$
= ____.**

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A mixed number must first become an improper fraction so I can flip and multiply.

Evidence: Students explain that the mixed number must be converted to an improper fraction ($7/2$) before applying Keep, Change, Flip.

Reasoning: This evidence connects the definition of divide mixed numbers to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can cut ____ pieces because $5\frac{1}{4} \div \frac{3}{4} = \frac{21}{4} \times \frac{4}{3} = \frac{84}{12} = \underline{\hspace{1cm}}$.**
- **My answer is ____.**

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Divide Mixed Numbers
2. **Notes 2:** Mixed Number
3. **Notes 3:** Improper Fraction
4. **Notes 4:** Convert
5. **Notes 5:** Simplify
6. **Notes 6:** Reciprocal
7. **Watch for this mistake:** *A common mistake in Divide Mixed Numbers is dividing the whole-number parts and fraction parts separately instead of converting first — for example, treating $4\frac{1}{2} \div 1\frac{1}{2}$ as $(4 \div 1) + (1/2 \div 1/2) = 5$. Always convert each mixed number to one improper fraction first, then divide using Keep, Change, Flip: $4\frac{1}{2} \div 1\frac{1}{2} = 9/2 \div 3/2 = 9/2 \times 2/3 = 3$.*
8. **Try It 1:** A. $2 - 3 \div 1\frac{1}{2} = 3 \div 3/2 = 3 \times 2/3 = 6/3 = 2$ sections.
9. **Try It 2:** A. $3 - 7\frac{1}{2} = 15/2$ and $2\frac{1}{2} = 5/2$. So $15/2 \div 5/2 = 15/2 \times 2/5 = 30/10 = 3$ doorways.